Ahmedabad University

Financial & Risk Analysis

Bayesian Generalized Pareto Modeling of Extreme Stock Market Crashes

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Project Context

Title : Bayesian Generalized Pareto Modeling of Extreme Stock Market Crashes

Domain : Finance and Risk Analysis

Real World Context :

- Normal-return assumption underestimates extreme losses
- Extreme events occur more often than expected

Why Uncertainty Is Intrinsic :

- Markets affected by global shocks and investor behavior
- Extreme crashes are rare and unpredictable

Project Objective and Scope

Project Objective : Estimate probability and severity of extreme market crashes

Project Scope :

- Focus exclusively on extreme market crashes
- Model tail risk using probabilistic methods (Bayesian GPD)
- Identify periods of elevated crash risk

Out of scope :

- Daily price prediction
- Assumptions of normal market behavior

Project System Overview

End to End Pipeline :

- Collect market price & volatility data
- Identify extreme loss events and model with Bayesian GPD
- Produce dynamic crash probability estimates

Major Components :

- Volatility computation module
- Bayesian regression model
- Crash probability output

Data Flow :

Market Data → Volatility → Extreme Events → Bayesian GPD Model → Risk Estimates

Sources of Uncertainty in the Project

Measurement Noise : Volatility and return estimates are noisy

Environmental / Contextual Variability : Global shocks, policy changes, investor behavior

Modeling / Data Uncertainty:

- Limited historical extreme events
- Simplifying Model Assumptions

Key Random Variables and Their Roles

Response Variable (Y):

- Extreme negative returns [Exceedances beyond −2% or −5%]
- Represents crash severity

Covariates (X):

- NIFTY 50 volatility: captures domestic market risk
- S&P 500 volatility: reflects global market influence
- Gold volatility: serves as a fear or safe-haven indicator

Probabilistic Models and Assumptions

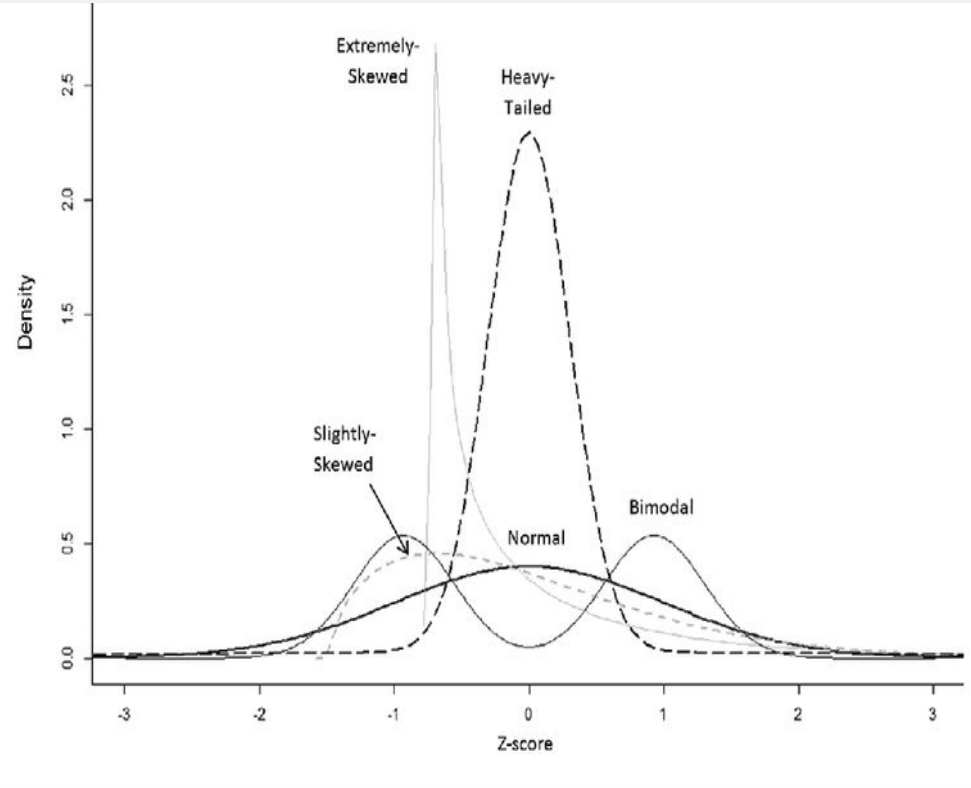
Model : Generalized Pareto Distribution (GPD) for extreme losses

Parameters :

1. Scale (σ): varies with volatility
2. Shape (ξ): tail heaviness (short-term constant)

Key Assumptions :

1. Extreme losses are heavy-tailed
2. Exceedances are conditionally independent
3. Volatility affects loss magnitude non-linearly



Probabilistic Reasoning and Inference Logic

Propagation of Uncertainty:

- As volatility changes, uncertainty increases.

Use of Probability Reasoning:

- Probabilistic outcome is obtained in models crash risk than a certain event.

Reasoning Pathway:

- Estimating the probability of market crash by using observed volatility so as to update the tail distribution.

Operationalizing Probabilistic Reasoning in the Project

Implementation of Reasoning:

- Various historical extremes are being trained under Bayesian GPD.

Evaluation Metric:

- RMSE is used to measure the accuracy of probabilistic estimates.

Simplifying Assumptions:

- Constant shape parameter
- Short-term market stability

Current Limitations and Conceptual Gaps

Known Weaknesses:

- Less observations are done for extreme crash.

Questionable Assumptions:

- Constant shape parameter
- Independent extreme events

Conceptual Ambiguities:

- Threshold selection
- Delayed global effects

Planned Refinements and Role Coordination

Next Milestone:

- Time-varying shape parameter

Tasks Ahead:

- Model lagged global volatility effects

Role Coordination:

- Model & data roles refine system
- Reviewer ensures consistency

Summary of Group Level Understanding

Probabilistic Understanding:

- Market crashes are very rare, major and heavy-tailed.
- Crash risk is being modeled as probability instead of predicting it.

Key Insight:

- Volatility doesn't cause market crashes but in case of high volatility chances of market crash increases.

Overall Conclusion:

- We use Bayesian modeling in order to capture crash risk.
- Global effects and time variation will be evolved by the model.

Thank You

Section 2 - Group 16

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